

## 1.11 Vector and matrix calculus

### 1.11.1 Gradients of linear and quadratic functions

For a differentiable scalar function  $f$  of  $\mathbf{x} \in \mathbb{R}^n$ , the **gradient**  $\partial f / \partial \mathbf{x}$  is the column vector with entries  $\partial f / \partial x_i$ , and the **Hessian**  $\partial^2 f / \partial \mathbf{x} \partial \mathbf{x}^\top$  is the  $n \times n$  matrix of second derivatives. For a vector-valued  $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$ , the  $m \times n$  *Jacobian*  $\partial \mathbf{g} / \partial \mathbf{x}^\top$  has  $(i, j)$  entry  $\partial g_i / \partial x_j$ .

#### Theorem 1.11.1 — Derivatives of linear and quadratic forms

Let  $\mathbf{a} \in \mathbb{R}^n$ , let  $\mathbf{A}$  be  $n \times n$  and  $\mathbf{B}$  be  $m \times n$ . Then

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{B}\mathbf{x})}{\partial \mathbf{x}^\top} = \mathbf{B}, \quad \frac{\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}, \quad \frac{\partial^2(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{A} + \mathbf{A}^\top.$$

In particular  $\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x}) / \partial \mathbf{x} = 2\mathbf{A}\mathbf{x}$  when  $\mathbf{A}$  is symmetric.

*Proof.* The first two are immediate from  $\mathbf{a}^\top \mathbf{x} = \sum_i a_i x_i$  and  $(\mathbf{B}\mathbf{x})_i = \sum_j b_{ij} x_j$ . For the quadratic form, expand

$$(\mathbf{x} + \mathbf{h})^\top \mathbf{A}(\mathbf{x} + \mathbf{h}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{h}^\top (\mathbf{A} + \mathbf{A}^\top)\mathbf{x} + \mathbf{h}^\top \mathbf{A}\mathbf{h}.$$

The middle term is linear in  $\mathbf{h}$  and the last is  $O(\|\mathbf{h}\|^2)$ , so  $(\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$  is the gradient. The gradient is linear in  $\mathbf{x}$  with matrix  $\mathbf{A} + \mathbf{A}^\top$ , which is the Hessian by the second formula. ■

#### Example. (The least squares criterion)

Let  $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{b} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\mathbf{b}$ . By Theorem 1.11.1,

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{b} = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}), \quad \frac{\partial^2 S}{\partial \mathbf{b} \partial \mathbf{b}^\top} = 2\mathbf{X}^\top \mathbf{X}.$$

Setting the gradient to zero gives the normal equations  $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ . The Hessian is nonnegative definite, so  $S$  is convex and every solution is a global minimizer. The next result proves this without calculus, and covers the rank-deficient case.

#### Proposition 1.11.2 — Minimizing a quadratic function

Let  $\mathbf{A}$  be  $n \times n$  nonnegative definite,  $\mathbf{b} \in \mathbb{R}^n$  and  $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$ . If  $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ , the minimizers of  $f$  are exactly the solutions of  $\mathbf{A}\mathbf{x} = \mathbf{b}$ , and the minimum value is  $-\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$  for any generalized inverse  $\mathbf{A}^-$ . If  $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$ , then  $f$  is unbounded below.

*Proof.* Let  $\mathbf{b} \in \mathcal{C}(\mathbf{A})$  and let  $\mathbf{x}_0$  solve  $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$ . Completing the square,

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{A}(\mathbf{x} - \mathbf{x}_0) - \mathbf{x}_0^\top \mathbf{A}\mathbf{x}_0.$$

The first term is nonnegative, and by Proposition 1.7.5(b) it is zero iff  $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$ , that is, iff  $\mathbf{Ax} = \mathbf{b}$ . The minimum is  $-\mathbf{x}_0^\top \mathbf{Ax}_0 = -\mathbf{x}_0^\top \mathbf{AA}^\top \mathbf{x}_0 = -\mathbf{b}^\top \mathbf{A}^\dagger \mathbf{b}$ . If  $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$ , write  $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$  with  $\mathbf{b}_1 \in \mathcal{C}(\mathbf{A})$  and  $\mathbf{0} \neq \mathbf{b}_2 \in \mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$  (Proposition 1.5.1). Then  $f(t\mathbf{b}_2) = -2t\|\mathbf{b}_2\|^2 \rightarrow -\infty$  as  $t \rightarrow \infty$ .

■

With  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ , the proposition gives least squares for every model matrix: the minimizers are the solutions of the normal equations, and the minimum of  $S$  is  $\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^\dagger \mathbf{X}^\top \mathbf{y}$ .

### 1.11.2 Constrained minimization

To minimize a differentiable  $f(\mathbf{x})$  subject to linear constraints  $\mathbf{Hx} = \mathbf{h}$ , with  $\mathbf{H}$  of size  $q \times n$ , the method of **Lagrange multipliers** looks for stationary points of  $f(\mathbf{x}) + 2\boldsymbol{\theta}^\top (\mathbf{Hx} - \mathbf{h})$  in  $(\mathbf{x}, \boldsymbol{\theta})$ . The factor 2 is only for convenience. For quadratic  $f$  the stationary point can be written down and checked directly.

#### Proposition 1.11.3

Let  $\mathbf{A}$  be positive definite,  $\mathbf{H}$  of size  $q \times n$  with full row rank, and  $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{Ax} - 2\mathbf{b}^\top \mathbf{x}$ . The unique minimizer of  $f$  subject to  $\mathbf{Hx} = \mathbf{h}$  is

$$\mathbf{x}_* = \mathbf{x}_0 - \mathbf{A}^{-1} \mathbf{H}^\top (\mathbf{HA}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{Hx}_0 - \mathbf{h}), \quad \mathbf{x}_0 = \mathbf{A}^{-1} \mathbf{b}.$$

*Proof.*  $\mathbf{HA}^{-1} \mathbf{H}^\top$  is positive definite by Proposition 1.7.5(c,e), so  $\mathbf{x}_*$  is well defined, and  $\mathbf{Hx}_* = \mathbf{h}$  by direct substitution. Stationarity of the Lagrangian gives  $2\mathbf{Ax}_* - 2\mathbf{b} + 2\mathbf{H}^\top \boldsymbol{\theta} = \mathbf{0}$ . Indeed  $\mathbf{Ax}_* - \mathbf{b} = -\mathbf{H}^\top \boldsymbol{\theta}_*$  with  $\boldsymbol{\theta}_* = (\mathbf{HA}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{Hx}_0 - \mathbf{h})$ . For any feasible  $\mathbf{x}$ , put  $\mathbf{d} = \mathbf{x} - \mathbf{x}_*$ , so that  $\mathbf{Hd} = \mathbf{0}$ . Then

$$f(\mathbf{x}) = f(\mathbf{x}_*) + 2\mathbf{d}^\top (\mathbf{Ax}_* - \mathbf{b}) + \mathbf{d}^\top \mathbf{Ad} = f(\mathbf{x}_*) - 2(\mathbf{Hd})^\top \boldsymbol{\theta}_* + \mathbf{d}^\top \mathbf{Ad} = f(\mathbf{x}_*) + \mathbf{d}^\top \mathbf{Ad},$$

which exceeds  $f(\mathbf{x}_*)$  unless  $\mathbf{d} = \mathbf{0}$ .

■

With  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ , this is least squares subject to a linear hypothesis  $\mathbf{H}\boldsymbol{\beta} = \mathbf{h}$ , the starting point of [Chapter 11](#).

### 1.11.3 Derivatives with respect to matrices and scalars

Likelihoods for normal models involve  $\log \det \boldsymbol{\Sigma}$  and  $\text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S})$ . For a scalar function  $f$  of an  $m \times n$  matrix  $\mathbf{W}$  with functionally independent entries,  $\partial f / \partial \mathbf{W}$  is the  $m \times n$  matrix with entries  $\partial f / \partial w_{ij}$ .

### Proposition 1.11.4

- (a)  $\partial \text{tr}(\mathbf{A}\mathbf{W})/\partial \mathbf{W} = \mathbf{A}^\top$ .
- (b) If  $\det \mathbf{W} > 0$ , then  $\partial \log \det \mathbf{W}/\partial \mathbf{W} = (\mathbf{W}^{-1})^\top$ .
- (c) If  $\mathbf{A}(t)$  is a nonsingular matrix whose entries are differentiable functions of a scalar  $t$ , with derivative  $\dot{\mathbf{A}} = \partial \mathbf{A}/\partial t$ , then

$$\frac{\partial \mathbf{A}^{-1}}{\partial t} = -\mathbf{A}^{-1} \dot{\mathbf{A}} \mathbf{A}^{-1}, \quad \frac{\partial \log |\det \mathbf{A}|}{\partial t} = \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}).$$

*Proof.*

- (a)  $\text{tr}(\mathbf{A}\mathbf{W}) = \sum_{i,j} a_{ji} w_{ij}$ . (c) Differentiate  $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$  by the product rule to get  $\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\partial \mathbf{A}^{-1}/\partial t = \mathbf{0}$ . For the determinant,  $\mathbf{A}(t+h) = \mathbf{A}(t)(\mathbf{I} + h\mathbf{A}^{-1}\dot{\mathbf{A}} + o(h))$ . In the permutation expansion of  $\det(\mathbf{I} + h\mathbf{C})$  only the diagonal product contributes terms of first order in  $h$ , so  $\det(\mathbf{I} + h\mathbf{C}) = 1 + h \text{tr} \mathbf{C} + O(h^2)$ . Hence  $\det \mathbf{A}(t+h) = \det \mathbf{A}(t)(1 + h \text{tr}(\mathbf{A}^{-1}\dot{\mathbf{A}}) + o(h))$ , and taking logarithms gives the formula. (b) Apply (c) with  $\mathbf{A}(t) = \mathbf{W} + t\mathbf{e}_i\mathbf{e}_j^\top$ . Then  $\text{tr}(\mathbf{W}^{-1}\mathbf{e}_i\mathbf{e}_j^\top) = \mathbf{e}_j^\top \mathbf{W}^{-1} \mathbf{e}_i$ , the  $(j, i)$  entry of  $\mathbf{W}^{-1}$ .

■

When  $\mathbf{W}$  is constrained to be symmetric,  $w_{ij}$  and  $w_{ji}$  are the same variable, so the derivatives with respect to off-diagonal entries double. Some books (Renchner and Schaalje 2008) report formulas of the type  $2\mathbf{W}^{-1} - \text{diag}(\mathbf{W}^{-1})$  for this reason. The simplest way to avoid the issue is to differentiate along a symmetric direction  $\mathbf{W} + t\mathbf{E}$ , as in (c). This is the convenient route for the normal likelihoods of [Chapter 32](#). The listing checks the gradient formulas against central finite differences, and the script also checks part (c) and [Proposition 1.11.2](#). The largest discrepancy in the listing is  $3 \times 10^{-9}$ , which is the size one expects from the finite-difference approximation.

```
import numpy as np
rng = np.random.default_rng(5)
p = 4
def num_grad(f, x, h=1e-6):
    """Central-difference gradient of a scalar function of a vector or matrix."""
    g = np.zeros_like(x)
    for idx in np.ndindex(x.shape):
        e = np.zeros_like(x)
        e[idx] = h
        g[idx] = (f(x + e) - f(x - e)) / (2 * h)
    return g

a = rng.normal(size=p)
A = rng.normal(size=(p, p)) # not symmetric
x = rng.normal(size=p)
checks = {
    "a'x": (num_grad(lambda v: a @ v, x), a),
    "x'Ax": (num_grad(lambda v: v @ A @ v, x), (A + A.T) @ x),
}
X = rng.normal(size=(20, p))
y = rng.normal(size=20)
b = rng.normal(size=p)
```

```

grad_ls = -2 * X.T @ (y - X @ b)
checks["||y - Xb||^2"] = (num_grad(lambda v: np.sum((y - X @ v) ** 2), b), grad_ls)

S = rng.normal(size=(p, p)); S = S @ S.T + p * np.eye(p) # positive definite
checks["log det S"] = (num_grad(lambda W: np.linalg.slogdet(W)[1], S), np.linalg.inv(S).T)
checks["tr(AS)"] = (num_grad(lambda W: np.trace(A @ W), S), A.T)
for name, (numeric, formula) in checks.items():
    print(f"{name:14s} max error {np.abs(numeric - formula).max():.1e}")

```

[Run this code online](#)

## 1.11.4 Exercises

### 1.11.4.1 A. Check your understanding

#### Exercise A1.

Observations fall in  $a$  clusters of size  $b$ , and  $\Sigma = \sigma^2 \mathbf{I}_a \otimes \mathbf{E}_b$ , where  $\mathbf{E}_b = (1 - \rho)\mathbf{I}_b + \rho\mathbf{J}_b$ . Find the eigenvalues, determinant and inverse of  $\Sigma$  using Proposition 1.10.1, Example (Equicorrelation) and Example (Eigenvalues of the equicorrelation matrix).

#### Exercise A2.

Use Proposition 1.11.2 to show that  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2$  with  $\lambda > 0$  has the unique minimizer  $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ , whatever the rank of  $\mathbf{X}$ .

### 1.11.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X}$  have full column rank,  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ , and let  $\mathbf{H}$  be  $q \times p$  with full row rank. Show that the minimizer of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\mathbf{H}\mathbf{b} = \mathbf{h}$  is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}), \quad \mathbf{W} = \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top,$$

and that the increase in the residual sum of squares is  $(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$ .

*Solution.* Apply Proposition 1.11.3 with  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ . Since  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} + f(\mathbf{b})$  and  $\mathbf{x}_0 = \hat{\boldsymbol{\beta}}$ , this gives  $\hat{\boldsymbol{\beta}}_H$ . By completing the square as in Proposition 1.11.2,  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$  for every  $\mathbf{b}$ . With  $\mathbf{b} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$ , the increase is

$$(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}) = (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}).$$

This quadratic form is the numerator of the  $F$  statistic in Chapter 11 (Theorem 11.3.2).

### 1.11.4.3 C. Going deeper

**Exercise C1.**

Let  $\mathbf{X}$  be  $n \times p$  of any rank, write  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ , and for  $\lambda > 0$  let  $\hat{\boldsymbol{\beta}}(\lambda) = (\mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$  be the minimizer of [Exercise \(A2\)](#). Show that

- (a)  $\frac{\partial \hat{\boldsymbol{\beta}}}{\partial \lambda} = -(\mathbf{A} + \lambda \mathbf{I})^{-1} \hat{\boldsymbol{\beta}}(\lambda)$ ;
- (b)  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}(\lambda)) = \lambda \hat{\boldsymbol{\beta}}(\lambda)$ ;
- (c)  $\|\hat{\boldsymbol{\beta}}(\lambda)\|^2$  is strictly decreasing in  $\lambda$ , and the residual sum of squares  $\|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}(\lambda)\|^2$  is strictly increasing, unless  $\mathbf{X}^\top \mathbf{y} = \mathbf{0}$ ;
- (d)  $\hat{\boldsymbol{\beta}}(\lambda) \rightarrow \mathbf{0}$  as  $\lambda \rightarrow \infty$ .

So the ridge path trades fit for size monotonically, which is what makes a single tuning parameter a sensible thing to choose ([Chapter 27](#)).

*Solution.*

- (a) By [Proposition 1.11.4\(c\)](#) with  $\mathbf{A}(\lambda) = \mathbf{A} + \lambda \mathbf{I}$  and  $\dot{\mathbf{A}} = \mathbf{I}$ , the derivative of  $(\mathbf{A} + \lambda \mathbf{I})^{-1}$  is  $-(\mathbf{A} + \lambda \mathbf{I})^{-2}$ , so  $\partial \hat{\boldsymbol{\beta}} / \partial \lambda = -(\mathbf{A} + \lambda \mathbf{I})^{-2} \mathbf{X}^\top \mathbf{y} = -(\mathbf{A} + \lambda \mathbf{I})^{-1} \hat{\boldsymbol{\beta}}$ .
- (b)  $\mathbf{X}^\top \mathbf{y} = (\mathbf{A} + \lambda \mathbf{I}) \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} + \lambda \hat{\boldsymbol{\beta}}$ , so  $\mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \lambda \hat{\boldsymbol{\beta}}$ .
- (c) Write  $\mathbf{K} = (\mathbf{A} + \lambda \mathbf{I})^{-1}$ , which is positive definite. Then

$$\frac{\partial}{\partial \lambda} \|\hat{\boldsymbol{\beta}}\|^2 = 2 \hat{\boldsymbol{\beta}}^\top \frac{\partial \hat{\boldsymbol{\beta}}}{\partial \lambda} = -2 \hat{\boldsymbol{\beta}}^\top \mathbf{K} \hat{\boldsymbol{\beta}} < 0$$

unless  $\hat{\boldsymbol{\beta}} = \mathbf{0}$ , which by (b) happens only when  $\mathbf{X}^\top \mathbf{y} = \mathbf{0}$ , and then  $\hat{\boldsymbol{\beta}}(\lambda) = \mathbf{0}$  for every  $\lambda$ . For the residual sum of squares, using (b),

$$\frac{\partial}{\partial \lambda} \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}\|^2 = -2(\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}})^\top \mathbf{X} \frac{\partial \hat{\boldsymbol{\beta}}}{\partial \lambda} = 2\lambda \hat{\boldsymbol{\beta}}^\top \mathbf{K} \hat{\boldsymbol{\beta}} > 0$$

under the same proviso.

- (d) Every eigenvalue of  $\mathbf{A} + \lambda \mathbf{I}$  is at least  $\lambda$ , so  $\|\mathbf{K}\|_2 \leq 1/\lambda$  and  $\|\hat{\boldsymbol{\beta}}(\lambda)\| \leq \|\mathbf{X}^\top \mathbf{y}\|/\lambda \rightarrow 0$ .

**Exercise C2.**

Show that  $\mathbf{W} \mapsto \log \det \mathbf{W}$  is strictly concave on the positive definite matrices: for positive definite  $\mathbf{W}$  and symmetric  $\mathbf{E} \neq \mathbf{0}$ , the function  $f(t) = \log \det(\mathbf{W} + t\mathbf{E})$  has  $f''(t) < 0$  wherever  $\mathbf{W} + t\mathbf{E}$  is positive definite. Deduce that  $\boldsymbol{\Omega} \mapsto \log \det \boldsymbol{\Omega} - \text{tr}(\boldsymbol{\Omega} \mathbf{S})$ , with  $\mathbf{S}$  positive definite, is strictly concave in the *concentration matrix*  $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$ , that its only stationary point is  $\boldsymbol{\Omega} = \mathbf{S}^{-1}$ , and hence that the maximum found in [Exercise \(C2\)](#) is unique. *Hint:* differentiate along symmetric directions, as the text recommends, and use [Theorem 1.7.6](#) and [Theorem 1.6.1](#).

*Solution.* Write  $\mathbf{M}(t) = (\mathbf{W} + t\mathbf{E})^{-1}$ . By [Proposition 1.11.4\(c\)](#),  $f'(t) = \text{tr}(\mathbf{M}\mathbf{E})$  and  $\mathbf{M}'(t) = -\mathbf{M}\mathbf{E}\mathbf{M}$ , so

$$f''(t) = \text{tr}(\mathbf{M}'\mathbf{E}) = -\text{tr}(\mathbf{M}\mathbf{E}\mathbf{M}\mathbf{E}) = -\text{tr}((\mathbf{M}^{1/2}\mathbf{E}\mathbf{M}^{1/2})^2) = -\|\mathbf{M}^{1/2}\mathbf{E}\mathbf{M}^{1/2}\|_F^2,$$

using the cyclic property to move one factor  $\mathbf{M}^{1/2}$  from the front to the back and  $\text{tr}(\mathbf{F}^2) = \text{tr}(\mathbf{F}^\top \mathbf{F}) = \|\mathbf{F}\|_F^2$  for symmetric  $\mathbf{F}$ . Since  $\mathbf{M}^{1/2}$  is nonsingular,  $\mathbf{M}^{1/2} \mathbf{E} \mathbf{M}^{1/2} = \mathbf{0}$  only for  $\mathbf{E} = \mathbf{0}$ , so  $f''(t) < 0$ . The positive definite matrices form a convex set, and strict concavity along every line segment in it is exactly strict concavity.

Adding the linear term  $-\text{tr}(\mathbf{\Omega} \mathbf{S})$  does not change the second derivative, so  $g(\mathbf{\Omega}) = \log \det \mathbf{\Omega} - \text{tr}(\mathbf{\Omega} \mathbf{S})$  is strictly concave. Differentiating along a symmetric direction  $\mathbf{E}$ ,

$$\left. \frac{\partial}{\partial t} g(\mathbf{\Omega} + t\mathbf{E}) \right|_{t=0} = \text{tr}(\mathbf{\Omega}^{-1} \mathbf{E}) - \text{tr}(\mathbf{S} \mathbf{E}) = \text{tr}\{(\mathbf{\Omega}^{-1} - \mathbf{S}) \mathbf{E}\}.$$

This vanishes for every symmetric  $\mathbf{E}$  iff  $\mathbf{\Omega}^{-1} = \mathbf{S}$ : take  $\mathbf{E} = \mathbf{\Omega}^{-1} - \mathbf{S}$ , which is symmetric, and the trace becomes  $\|\mathbf{\Omega}^{-1} - \mathbf{S}\|_F^2$ . A strictly concave function has at most one stationary point and no other local maximum, so  $\mathbf{\Omega} = \mathbf{S}^{-1}$ , that is,  $\mathbf{\Sigma} = \mathbf{S}$ , is the unique maximizer, as Exercise (C2) found by a different route.